



StudyStream

Intelligent Student Planning Platform

Stop wondering what you're forgetting.

Django 6

Python 3.11+

SQLite

REST API

Group 3 · Haylee · Teresa · Kaheel · Aryan



Students are Drowning

Full-Time Student

4–6

Courses juggled
per semester

Overlapping deadlines every week

Working Student

60%

of working students
miss deadlines

20h work/week breaks any plan

Exec. Dysfunction

48h

avg panic window
before crisis

No forecasting — lists fail



A Full Planning Engine

01 Subtask Breakdown

Assignments auto-split into ordered, trackable steps

02 Time-Block Scheduling

Study blocks auto-placed based on available free time

03 Workload Forecasting

GREEN / YELLOW / RED signal keeps overload visible

04 Conflict Detection

Auto-detects clashes and surfaces smart suggestions

05 Progress Analytics

Track completion rates, trends, and projected outcomes

06 Secure Auth & Profiles

Per-user preferences: sleep hours, commute, semester



Feature Walkthrough

Workload Status

Class + work + assignment hours vs. available hours. Real-time per week.

Recurring Events

Weekly, biweekly, monthly event patterns auto-mapped into schedule.

Subtask Cascade

save() triggers rollup — parent completion auto-calculated from children.

workload_config.py

```
WEEK_HOURS = 168
```

```
SLEEP_HOURS = 49
```

```
if utilization >= 0.85
```

```
    status = "RED"
```

```
elif utilization >= 0.65
```

```
    status = "YELLOW"
```

```
else:
```

```
    status = "GREEN"
```



GREEN



YELLOW



RED



Database Models

User / Profile

username · sleep_hours ·
commute_hours

Semester

name · start_date · end_date ·
is_active

Course

code · color_hex ·
weekly_study_hours

Assignment

title · due_date · status ·
priority · estimated_hours

AssignmentSubtask

step_order · is_complete →
triggers rollup

WorkloadAnalysis

utilization_ratio · week_status
(G/Y/R) · recommended_actions[]



Three-Layer Architecture

PRESENTATION

Django Templates · Dashboard Widgets · AJAX UI

Dynamic assignment cards, real-time workload meters, interactive timeline — all rendered via Django templating with JavaScript/AJAX polling for live updates.

APPLICATION

accounts/ · home/ · core/

accounts/ — auth, courses, profiles · home/ — workload engine, forecasting · core/ — conflict detection, ScheduledItem API, event scheduling.

DATA

SQLite → PostgreSQL-ready · Django ORM · UUID PKs

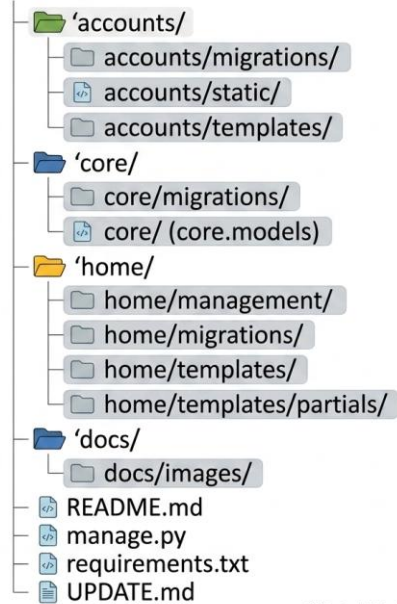
WorkloadAnalysis per user/week · migration-safe schema · python-dotenv environment config · REST endpoints at /accounts/api/* and /home/api/*



Project Structure

Folder Structure

./cts-289-group3-capstone_project_studystream/



*(Portrait Mode: Vertical Summary - Simplified Tree)

scheduling/conflict.py

```
@dataclass(frozen=True)
class ScheduleConflict:
    requested_item: ScheduledItem
    conflicting_item: ScheduledItem
    suggestion_item: ScheduledItem | None
    message: str

    @property
    def replaceable(self) -> bool:
        return (
            self.object_id is not None
            and self.local_ref is None
        )
```

Technologies & Toolchain

BE

Backend

- Django 6.0.4
- Python 3.11+
- SQLite / ORM
- UUID primary keys
- python-dotenv

FE

Frontend

- Django Templates
- JavaScript / AJAX
- CSS Variables
- 8-color HEX palette
- Responsive layout

DT

Dev Tools

- GitHub Codespaces
- manage.py CLI
- seed_demo_data
- run_workload_analysis
- REST API testing



Four-Phase Build Plan

Phase 01

Foundation

Auth system

Semester & Course DB

Task model scaffold

Phase 02

Core Engine

Subtask breakdown

Time-block scheduling

Conflict detection

Phase 03

Intelligence

Workload G/Y/R engine

Deadline cluster alerts

Email notifications

Phase 04

Expansion

Group collaboration

UI refinement

Recurring events



Future Improvements

MOB

Native Mobile App

React Native or Flutter

GRP

Group Collaboration

Shared workspaces, team calendars

AI

AI Scheduling Advisor

LLM-powered study optimization

LMS

LMS Integration

Canvas & Blackboard API sync

ANL

Advanced Analytics

GPA trends, CSV export, forecasts

NTF

Smart Notifications

Adaptive deadline-aware alerts

Thank you
Questions?

Group 3 · Haylee · Teresa · Kaheel · Aryan